CSE4076 · IMAGE AND VIDEO ANALYTICS

Real-Time Illicit Activity Detection System

Video processing and analytics for detecting fighting, robbery and vandalism in CCTV footage



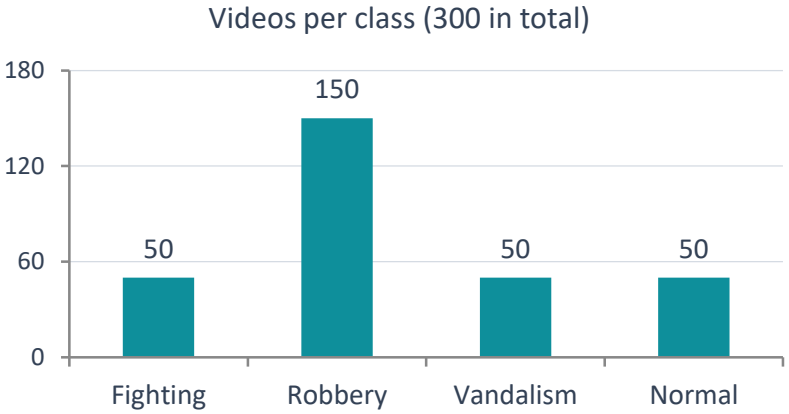
Project presentation to the faculty

Introduction: why automatic CCTV analysis

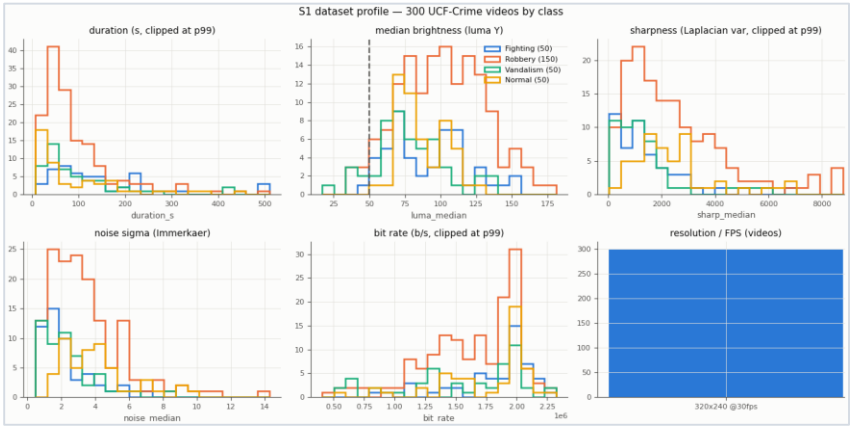
Problem	Watching many CCTV feeds by hand does not scale, and incidents are noticed late
Challenge	Low-light, 320 × 240, heavily compressed footage with timestamps, bars and camera cuts
Objective	Classify Fighting, Robbery, Vandalism or Normal in real time and raise an alert with evidence
This work	Make the video processing and analytics before training explicit, measurable and visible



Dataset: UCF-Crime, four classes



- Official 4-fold Action_Recognition split, with a train/test leakage check
- All videos 30 FPS, 320 × 240; 206 have letterbox bars, 218 a timestamp or logo
- Temporal ground truth for the test videos



Profile of all 300 videos by class

Supporting data: ExDARK (7,363 dark images) and LLVIP (15,488 night images) train Zero-DCE++; LLVIP + ExDARK People (12,634) fine-tune YOLOv11s.

Proposed methodology: overview

S1
Profile
PyAV stats, thresholds

S2
Resample
exactly 12 FPS

S3
Clean
cuts, bars, overlay, frozen

S4
Enhance
Zero-DCE++, denoise, deflicker

S5
Camera comp.
skipped: 2 / 300 videos move

S6
Motion
MOG2, optical flow, MHI

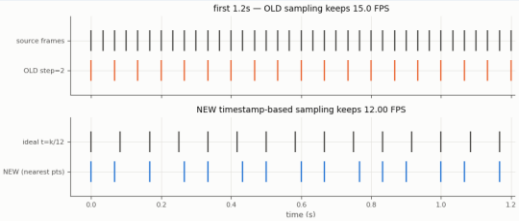
S7
Actors
YOLO, ByteTrack, crop

S8
Select
activity, proposals, gate

S9
Package
uint8 clips, sidecars

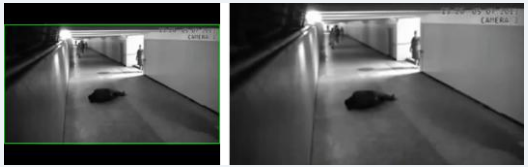
Principles. No training here: Zero-DCE++ and YOLO weights are reused, RAFT-small, FastDVDnet and TransNetV2 are pretrained, the rest is classical. Every operation saves a sample input and output.

Methodology: consistent, clean, enhanced frames



S2 Exact 12 FPS

Pick the frame nearest to t = k/12 s. The old step rule gave 15 FPS.



S3 Structural cleaning

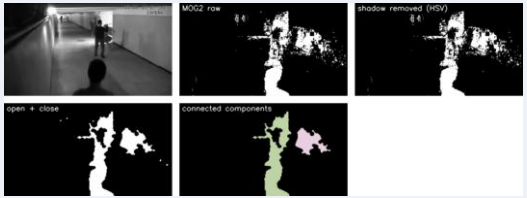
Crop bars, inpaint timestamps, detect cuts (HSV + edges, TransNetV2), flag frozen frames.



S4 Enhancement

One luma threshold, Zero-DCE++, FastDVDnet temporal denoise, brightness deflicker.

Methodology: motion, actors and clip selection



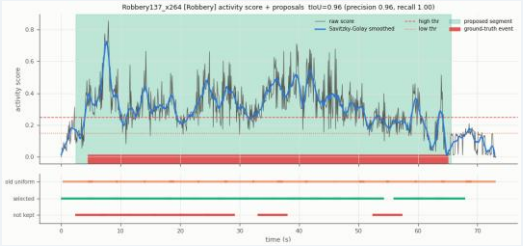
S6 Motion analysis

MOG2 with HSV shadow removal, Farneback and RAFT-small flow, entropy, MHI.



S7 Actor analysis

Detection, ByteTrack IDs, interaction signals, smoothed actor-centred 224 crop.



S8 Clip selection

Activity score and action proposals; clips inside proposals; six-rule quality gate.

System architecture: from camera to alert

Input	Processing	Inference	Output
CCTV stream	**Zero-DCE++**	**Redis Streams**	**Supabase storage + database**
RTSP / mp4 / webcam	if luma below 50	VideoMAE worker	Twilio SMS / MMS
Frames at 12 FPS	YOLOv11s + ByteTrack	Fighting / Robbery / Vandalism / Normal	React live dashboard
	16-frame clip buffer		

Offline training path: UCF-Crime videos → S1 to S9 video processing and analytics → 16-frame clips → VideoMAE fine-tuning (4-fold CV) used by the inference worker

Results: effect of the video processing

15 → 12.00

effective clip FPS

8 / 8

demo videos with lower noise after denoising

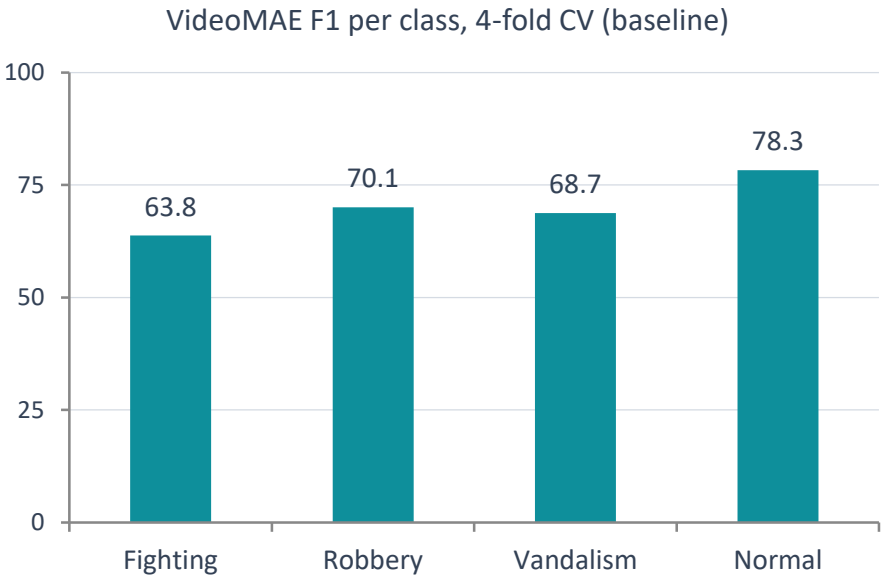
0.96

best tIoU of proposals vs ground truth (0.35, 0.15 on the others)

Operation	Before	After
Framing	Stretched 640 × 640	**Letterbox + actor crop**
Brightness	Three different checks	**One luma rule, threshold 50**
Sampling	Uniform, label noise	**Inside proposals**
Validity	std > 0.01	**Six-rule quality gate**
Tracking	None committed	**ByteTrack + signals**
Clip size	9.6 MB float32	**2.4 MB uint8**

Measured on the fixed 8-video sample set; every operation has a saved input/output sample.

Results: detection and action recognition



Overall macro-F1 70.2 ± 5.6 %. Baseline from the old clips; retraining on the new clips is next.

0.907

YOLOv11s mAP50 (mAP50-95 0.534)

2.0 ms

per frame; precision 0.869, recall 0.850

Finding. On UCF-Crime frames 34 % of the fine-tuned boxes are frame-sized, so COCO yolo11s.pt is used for tracking and cropping.

Conclusion

1. A complete video processing and analytics pipeline, needing no training, now prepares every clip, with a saved input and output sample for each operation.

2. Five preprocessing gaps are fixed: frame rate, aspect ratio, brightness rule, clip sampling and clip validation.

3. Next: extract all four folds, retrain VideoMAE, and run ablations on sampling, framing, enhancement and cleaning.

Thank you. Questions?